

**SIH26070**

# **CycloVision AI**

## **An AI/ML-Based System for Identification, Classification and Prediction of Tropical Cyclone Patterns Using Multi-Source Satellite Data**

**Problem Statement ID:** SIH26070

**Organization:** Ministry of Earth Sciences (MoES) / India Meteorological Department (IMD)

**Category:** Software

**Theme:** Disaster Management

This document presents a comprehensive research and development concept for an Artificial Intelligence and Machine Learning based system capable of identifying, classifying, tracking and predicting tropical cyclone patterns using multi-source satellite, atmospheric and oceanographic data.

# 1. Introduction

Tropical cyclones are among the most destructive natural disasters. They can cause heavy rainfall, strong winds, storm surges, coastal flooding, infrastructure damage, agricultural losses and disruption of transportation and communication. India is particularly vulnerable because of its long coastline and exposure to cyclones originating in the Bay of Bengal and Arabian Sea.

Traditional cyclone monitoring depends on meteorological experts, satellite interpretation, radar observations, numerical weather prediction models and ground observations. The proposed SIH solution uses Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, GIS and multi-source data fusion to automate and improve cyclone monitoring.

# 2. Main Purpose of the Project

- Automatically detect cyclones from satellite images.
- Identify cyclone location and movement.
- Classify cyclone intensity and development stage.
- Recognize different cyclone patterns.
- Track cyclone movement over time.
- Predict future path and possible landfall.
- Estimate intensity strengthening or weakening.
- Generate early warnings.
- Display results through an interactive dashboard.
- Assist meteorologists and disaster-management authorities.

# 3. Proposed Project Title

**CycloVision AI**

*An Intelligent Multi-Source Satellite Data Fusion System for Tropical Cyclone Detection, Classification and Prediction*

Other possible names: CycloneNet AI, StormSense, CycloTrack AI, CycloPredict, AeroStorm Intelligence System, Cyclone Sentinel and TropiScan AI.

**Recommended tagline:** Detect. Understand. Predict. Protect.

# 4. What is a Tropical Cyclone?

A tropical cyclone is a large rotating storm system that forms over warm tropical oceans. It generally consists of a low-pressure centre, strong rotating winds, spiral cloud bands, heavy rainfall, an eye, eyewall and dense cloud formation. Cyclones obtain much of their energy from warm ocean water.

# 5. Tropical Cyclone Life Cycle

| Stage                 | Description                                                  |
|-----------------------|--------------------------------------------------------------|
| Tropical Disturbance  | Initial atmospheric disturbance and cloud formation.         |
| Tropical Depression   | Low-pressure system begins developing organized circulation. |
| Deep Depression       | Wind speed and cloud organization increase.                  |
| Cyclonic Storm        | Well-defined cyclonic circulation develops.                  |
| Severe Cyclonic Storm | Storm becomes more intense with stronger winds.              |

| Stage                           | Description                                                             |
|---------------------------------|-------------------------------------------------------------------------|
| Very Severe Cyclonic Storm      | Highly organized cloud structure and dangerous wind conditions.         |
| Extremely Severe Cyclonic Storm | Very intense storm with significant destructive potential.              |
| Super Cyclonic Storm            | Highest category of cyclone intensity in the IMD classification system. |
| Landfall                        | Cyclone crosses the coastline.                                          |
| Dissipation                     | Cyclone gradually loses energy.                                         |

## 6. Core Objectives

| Objective            | Description                              |
|----------------------|------------------------------------------|
| Detection            | Detect cyclone presence automatically.   |
| Classification       | Identify cyclone category and intensity. |
| Pattern Recognition  | Recognize cloud and spiral structures.   |
| Tracking             | Track cyclone movement.                  |
| Prediction           | Predict future location and path.        |
| Intensity Prediction | Estimate strengthening or weakening.     |
| Early Warning        | Alert authorities and citizens.          |
| Visualization        | Show results on maps and dashboards.     |
| Decision Support     | Help disaster-management agencies.       |

## 7. Proposed System Architecture

Multi-Source Data → Data Collection Layer → Data Preprocessing → Multi-Source Data Fusion → AI/ML Engine → Prediction Engine → Dashboard and Alert System

- Data collection: satellite images, weather data, ocean data, historical cyclone data, radar data and numerical weather prediction data.
- Preprocessing: cleaning, image enhancement, resizing, normalization, geo-referencing and synchronization.
- AI/ML engine: cyclone detection, classification, pattern recognition, path prediction and intensity prediction.
- Final outputs: interactive maps, cyclone tracks, risk zones, forecasts and early warnings.

## 8. Data Sources to Be Used

### A. Satellite Data

**Indian Satellites:** INSAT-3D, INSAT-3DR and INSAT-3DS can provide meteorological observations useful for cloud monitoring, storm development, temperature and atmospheric analysis.

Possible data types include visible imagery, infrared imagery, water vapour imagery, thermal infrared data, atmospheric temperature and humidity profiles.

### B. MOSDAC

The Meteorological and Oceanographic Satellite Data Archival Centre (MOSDAC) is a major Indian source for meteorological and oceanographic satellite products, observations and geospatial services. Useful formats can include NetCDF, GeoTIFF, HDF and image products.

### C. International Sources

- NOAA: sea surface temperature, atmospheric data, infrared imagery and historical storm information.
- NASA Earth observation sources: cloud data, ocean temperature and atmospheric measurements.
- Himawari satellite: high-frequency weather monitoring, cloud movement and tropical cyclone observation.

## 9. Other Data Sources

| Source                       | Useful Information                                                                     |
|------------------------------|----------------------------------------------------------------------------------------|
| Weather Data                 | Air temperature, pressure, humidity, wind speed, wind direction and rainfall.          |
| Ocean Data                   | Sea surface temperature, ocean heat content, wave height, pressure and ocean currents. |
| Radar Data                   | Rainfall intensity, storm structure, wind and cloud movement.                          |
| Historical Cyclone Data      | Cyclone name, date, latitude, longitude, wind speed, pressure, category and landfall.  |
| Numerical Weather Prediction | Future atmospheric conditions, wind, temperature and pressure forecasts.               |

## 10. Multi-Source Data Fusion

The main innovation is combining different sources instead of relying on one satellite image. Satellite data provides cloud structure; infrared data provides cloud temperature; radar provides rainfall and storm structure; ocean data provides sea conditions; weather stations provide ground observations; historical records provide previous patterns; and NWP models provide atmospheric forecasts.

**Concept:** Satellite + Weather + Ocean + Historical Tracks + Radar → Data Fusion → AI/ML Model → Cyclone Analysis.

## 11. Main Technologies Required

**Primary programming language:** Python

Python is recommended because of its strong ecosystem for AI, machine learning, scientific computing, satellite-data processing and visualization.

| Category         | Technologies                    |
|------------------|---------------------------------|
| Data Processing  | NumPy, Pandas, SciPy            |
| Machine Learning | Scikit-learn                    |
| Deep Learning    | TensorFlow or PyTorch           |
| Image Processing | OpenCV, Pillow                  |
| Satellite Data   | Xarray, NetCDF4, GDAL, Rasterio |
| Visualization    | Matplotlib, Plotly              |
| GIS              | GeoPandas, Folium, Leaflet      |

## 12. AI Models

### Module 1: Cyclone Detection

CNNs can learn cyclone-specific visual features such as spiral clouds, cyclone eyes, dense cloud areas and rain bands. YOLO can be used for near-real-time object detection, providing cyclone location, bounding boxes and confidence

scores.

### Module 2: Cyclone Classification

Recommended models include CNN, EfficientNet or ResNet. The classifier can distinguish no cyclone, tropical disturbance, depression, deep depression and different cyclone intensity categories.

### Module 3: Pattern Recognition

- Spiral pattern
- Eye formation
- Circular cloud structure
- Asymmetric cloud formation
- Rapid intensification
- Weakening pattern
- Rain-band expansion

### Module 4: Path Prediction

LSTM and GRU models are suitable for time-series prediction using latitude, longitude, time, wind speed, pressure, sea surface temperature, wind direction, humidity and previous tracks. Transformer models can be explored for advanced long-term multi-variable sequence modelling.

### Module 5: Intensity Prediction

Inputs may include sea surface temperature, wind speed, atmospheric pressure, humidity, cloud temperature and ocean heat content. The model predicts strengthening, weakening or stable intensity.

## 13. Recommended Hybrid AI Architecture

Satellite Image → CNN/YOLO → Cyclone Detection → Feature Extraction + Weather Data + Ocean Data → Data Fusion → LSTM/GRU → Path Prediction → Intensity Prediction.

## 14. Project Modules

| Module                      | Function                                                                            |
|-----------------------------|-------------------------------------------------------------------------------------|
| 1. Data Acquisition         | Collect satellite, weather, ocean and historical cyclone data.                      |
| 2. Data Preprocessing       | Cleaning, missing-value treatment, resizing, normalization and timestamp alignment. |
| 3. Cyclone Detection        | Identify cyclone presence and centre location.                                      |
| 4. Classification           | Determine category, stage and intensity.                                            |
| 5. Pattern Analysis         | Analyse spiral formation, eye structure and cloud density.                          |
| 6. Path Prediction          | Predict future latitude, longitude, direction and speed.                            |
| 7. Intensity Prediction     | Predict wind speed, pressure and category changes.                                  |
| 8. Risk Assessment          | Identify high-risk areas and possible landfall regions.                             |
| 9. Alert Generation         | Generate green, yellow, orange and red alerts.                                      |
| 10. Visualization Dashboard | Display satellite images, tracks, forecasts and risk zones.                         |

## 15. Alert Levels

| Level  | Meaning                                    |
|--------|--------------------------------------------|
| Green  | Low risk.                                  |
| Yellow | Watch and monitor.                         |
| Orange | Prepare for severe conditions.             |
| Red    | Immediate danger and maximum preparedness. |

## 16. Frontend, Backend and Database

| Layer             | Recommended Technology  |
|-------------------|-------------------------|
| Frontend          | React.js + Tailwind CSS |
| Backend           | FastAPI                 |
| Programming       | Python                  |
| Deep Learning     | PyTorch                 |
| Computer Vision   | OpenCV                  |
| Detection         | YOLO                    |
| Classification    | EfficientNet            |
| Prediction        | LSTM / GRU              |
| Database          | PostgreSQL + PostGIS    |
| GIS Visualization | Leaflet / Mapbox        |
| Deployment        | Docker + Cloud          |

## 17. Materials and Resources Required

### Hardware

- Laptop or desktop computer.
- Minimum 8 GB RAM; 16 GB recommended.
- Multi-core processor.
- Optional NVIDIA GPU for deep-learning training.
- Cloud alternatives such as Google Colab or Kaggle GPU.

### Software

- Python
- Jupyter Notebook
- VS Code
- GitHub
- PostgreSQL
- Docker
- Node.js

## 18. Dataset Requirements

The project dataset should include three major categories.

## A. Satellite Images

- Visible images
- Infrared images
- Water vapour images

## B. Numerical Weather Data

- Date and time
- Latitude and longitude
- Pressure
- Wind speed
- Humidity
- Temperature

## C. Historical Cyclone Track Data

- Cyclone name
- Timestamp
- Latitude
- Longitude
- Wind speed
- Pressure
- Category

# 19. Suggested Dataset Structure

```
cyclone_dataset/  
■  
■■■ satellite_images/  
■ ■■■ cyclone/  
■ ■■■ non_cyclone/  
■■■ weather_data/  
■ ■■■ weather.csv  
■■■ ocean_data/  
■ ■■■ ocean.csv  
■■■ historical_tracks/  
■ ■■■ cyclone_tracks.csv  
■■■ processed_data/
```

# 20. Data Preprocessing

## Satellite Images

- Noise removal
- Resizing
- Normalization
- Cloud enhancement
- Feature extraction

## Numerical Data

- Missing-value treatment
- Data cleaning
- Normalization
- Feature selection
- Training-ready formatting

## 21. Feature Engineering

| Feature Type | Examples                                                                 |
|--------------|--------------------------------------------------------------------------|
| Atmospheric  | Pressure, temperature, humidity, wind speed and direction.               |
| Ocean        | Sea surface temperature, ocean heat content and wave height.             |
| Satellite    | Cloud brightness, cloud temperature, spiral structure and eye formation. |
| Temporal     | Time, date, season and previous cyclone location.                        |

## 22. Prediction Outputs

- Current cyclone location.
- Cyclone intensity.
- Cyclone category.
- Movement direction.
- Predicted path.
- Possible landfall area.
- Risk level.
- Prediction confidence score.

## 23. User Interface Design

| Page                | Features                                                    |
|---------------------|-------------------------------------------------------------|
| Home Page           | Project overview, current weather and active cyclone count. |
| Live Monitoring     | Satellite image and real-time detection.                    |
| Prediction Page     | Predicted track and future positions.                       |
| Risk Map            | Coastal regions and risk zones.                             |
| Historical Analysis | Previous cyclones and comparison graphs.                    |

## 24. Applications

- Disaster management and evacuation planning.
- Early warning systems.
- Coastal safety.
- Aviation weather monitoring.

- Marine transportation safety.
- Fishermen safety alerts.
- Agricultural preparedness.
- Insurance risk estimation.
- Scientific and climate research.

## 25. Advantages

- Faster analysis of large image volumes.
- Automated and continuous monitoring.
- Potentially improved decisions through multi-source fusion.
- Earlier warning and preparation time.
- Reduced manual workload.
- Near-real-time analysis.
- Large ocean-area coverage.
- Learning from historical cyclone patterns.
- Interactive visualization.
- Scalable to floods, rainfall, thunderstorms and heatwaves.

## 26. Disadvantages and Limitations

- Real-time data availability may be limited.
- Satellite data volume can be very large.
- Deep learning may require significant computing power.
- Weather prediction inherently contains uncertainty.
- Images may contain noise or missing observations.
- Model quality depends on training data.
- Real-time deployment requires reliable internet.
- Deep-learning models can be difficult to interpret.

## 27. Key Innovations for SIH

### Innovation 1: Multi-Source Fusion Engine

Combines satellite, weather, ocean, historical and radar data into a unified AI decision pipeline.

### Innovation 2: Explainable AI

The system can explain why a prediction was generated, for example: high sea surface temperature, decreasing central pressure, increasing wind speed and stronger cloud organization.

### Innovation 3: Cyclone Pattern Similarity Engine

Compare the current cyclone with historical cyclones and calculate similarity scores to identify past events with comparable behaviour.

### Innovation 4: AI Confidence Score

Display confidence separately for path prediction, intensity prediction and landfall estimation.

### Innovation 5: Rapid Intensification Detection

Monitor sudden changes in wind, pressure and cloud organization to identify possible rapid strengthening.

### Innovation 6: Digital Twin Concept

Create a simplified dynamic representation of current cyclone position, cloud movement, future path and intensity changes.

### Innovation 7: Risk-Based Impact Prediction

Extend the system beyond weather prediction by estimating landfall region, population exposure, infrastructure risk and flood risk.

## 28. Final Solution Architecture

Data Sources → Data Pipeline → Cleaning and Alignment → Normalization → Feature Extraction → Multi-Source Data Fusion → YOLO/CNN Detection + EfficientNet Classification + LSTM Prediction → AI Decision Engine → Path Forecast + Intensity + Risk Analysis → FastAPI → Database → Web Dashboard → Alert System.

## 29. Development Roadmap

| Phase | Work                                                                        |
|-------|-----------------------------------------------------------------------------|
| 1     | Research tropical cyclones, satellite data and existing prediction methods. |
| 2     | Collect satellite images, historical tracks, weather and ocean datasets.    |
| 3     | Preprocess and prepare datasets.                                            |
| 4     | Develop cyclone detection model.                                            |
| 5     | Develop classification model.                                               |
| 6     | Develop LSTM/GRU prediction model.                                          |
| 7     | Integrate all AI modules.                                                   |
| 8     | Develop dashboard.                                                          |
| 9     | Test using historical cyclones.                                             |
| 10    | Deploy prototype.                                                           |

## 30. Evaluation Metrics

| Task                 | Metrics                                           |
|----------------------|---------------------------------------------------|
| Detection            | Accuracy, Precision, Recall, F1 Score and mAP.    |
| Classification       | Accuracy, Confusion Matrix, Precision and Recall. |
| Path Prediction      | MAE, RMSE and geographical distance error.        |
| Intensity Prediction | MAE and RMSE.                                     |

## 31. Suggested Team Division

| Team Member | Responsibility                       |
|-------------|--------------------------------------|
| Member 1    | Team lead and research coordination. |
| Member 2    | Data collection and preprocessing.   |
| Member 3    | Machine learning model development.  |
| Member 4    | Deep learning and computer vision.   |
| Member 5    | Frontend and dashboard.              |
| Member 6    | Backend and database.                |

### 32. Minimum Viable Product (MVP)

- Upload or access a satellite image.
- Automatically detect cyclone presence.
- Classify cyclone intensity.
- Predict next locations from historical sequences.
- Display results on an interactive GIS dashboard.

### 33. Future Scope

- Flood prediction
- Storm surge prediction
- Rainfall prediction
- Tsunami integration
- Mobile application
- SMS alerts
- Regional-language alerts
- Voice-based warnings
- Drone-data integration
- IoT weather sensors
- Climate-change analysis

### 34. Problem–Solution Summary

| Problem                        | Proposed Solution                  |
|--------------------------------|------------------------------------|
| Manual satellite analysis      | AI-based automated detection.      |
| Disconnected data sources      | Multi-source data fusion.          |
| Delayed pattern recognition    | Automated near-real-time analysis. |
| Difficult cyclone tracking     | AI trajectory prediction.          |
| Uncertain intensity changes    | ML intensity prediction.           |
| Complex scientific information | Simple GIS dashboard.              |
| Limited preparedness           | Risk-based early alerts.           |

## 35. Final Project Concept

**CycloVision AI** is a web-based Artificial Intelligence and Machine Learning platform designed to monitor tropical cyclones using multi-source satellite and meteorological data. The system integrates satellite imagery, weather parameters, ocean conditions, historical cyclone records and radar observations where available.

Computer-vision models automatically identify cyclonic structures from satellite images. Classification models determine cyclone intensity and development stage. Time-series deep-learning models analyse movement and atmospheric conditions to predict future trajectory and intensity.

The final output is presented through an interactive GIS-based dashboard showing cyclone location, predicted track, intensity, confidence level, risk zones and early-warning alerts.

**Unique Selling Point:** CycloVision AI is not just a cyclone detection system. It is an intelligent multi-source disaster decision-support platform that combines satellite imagery, atmospheric observations, ocean conditions and historical cyclone behaviour to detect, classify, track, predict and assess cyclone risks in near real time.

## 36. Recommended Final SIH Implementation

- YOLO for cyclone detection.
- EfficientNet for intensity and pattern classification.
- LSTM or GRU for path prediction.
- INSAT/MOSDAC and other public satellite data sources.
- Historical cyclone tracks and weather/ocean parameters.
- Python and FastAPI backend.
- React-based GIS dashboard.
- Historical cyclone similarity engine.
- Explainable AI outputs.
- Rapid intensification alerts.
- Risk-based landfall impact visualization.

## References and Suggested Data Portals

Smart India Hackathon problem statement portal: SIH26070.

MOSDAC – Meteorological and Oceanographic Satellite Data Archival Centre.

INSAT meteorological satellite data and products.

India Meteorological Department cyclone information and classifications.

NOAA and NASA public Earth observation datasets.